

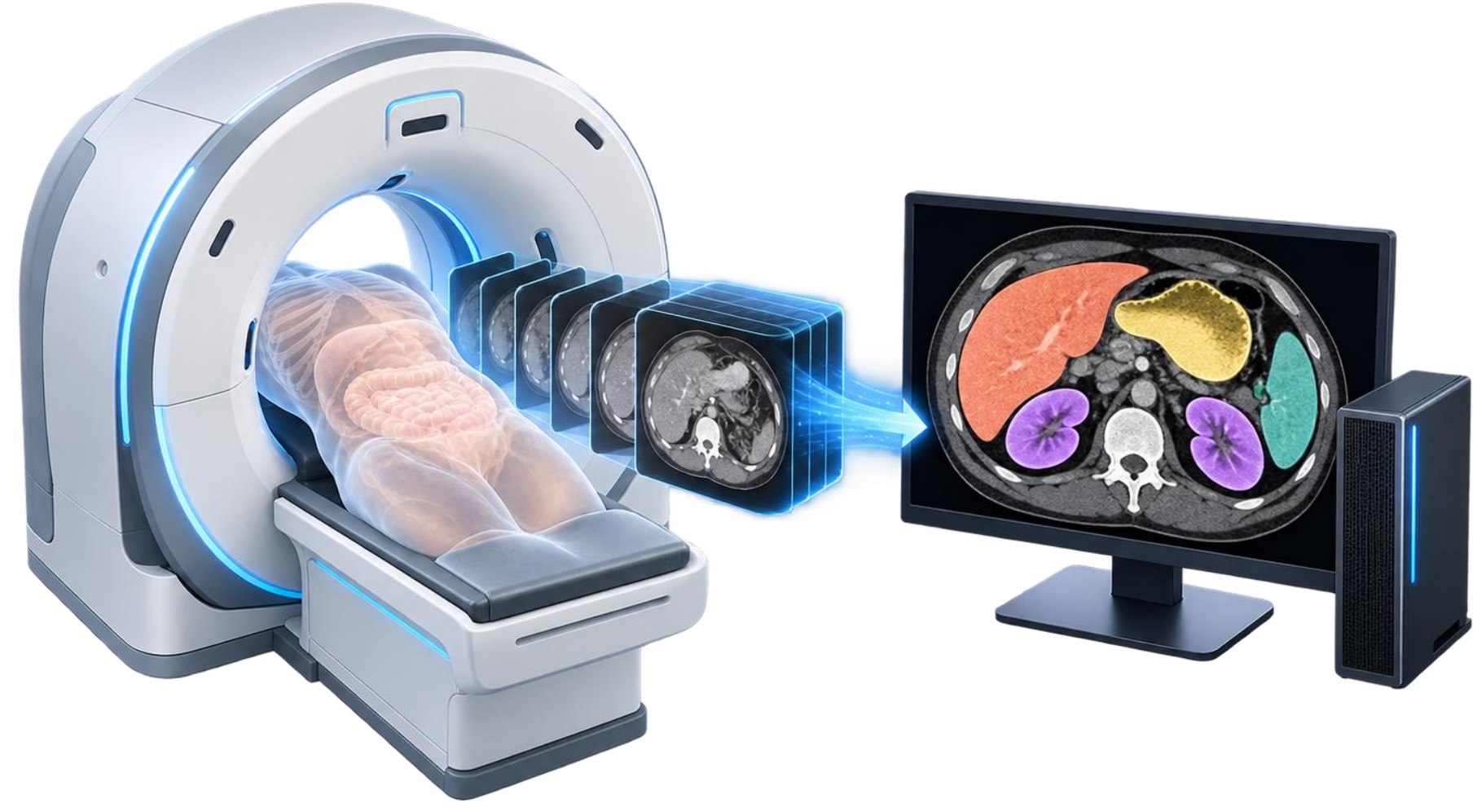
B-MIM: Biased Masked Image Modeling for Generalizable Segmentation of Fine-Grained Anatomical Structures

Sebastián González¹, Karen Sanchez², José M. Saavedra³, Marcelo Pizarro³, Bernard Ghanem²

¹Universidad de Chile, Santiago, Chile; ²King Abdullah University of Science and Technology (KAUST), Saudi Arabia; ³Universidad de los Andes, Santiago, Chile.

1. Introduction

Current CT encoders are biased towards coarse semantics.



Goal:

Learn transferable CT representations that better capture **fine-grained** anatomical details and generalize across datasets and tasks.

3. Datasets & Evaluation

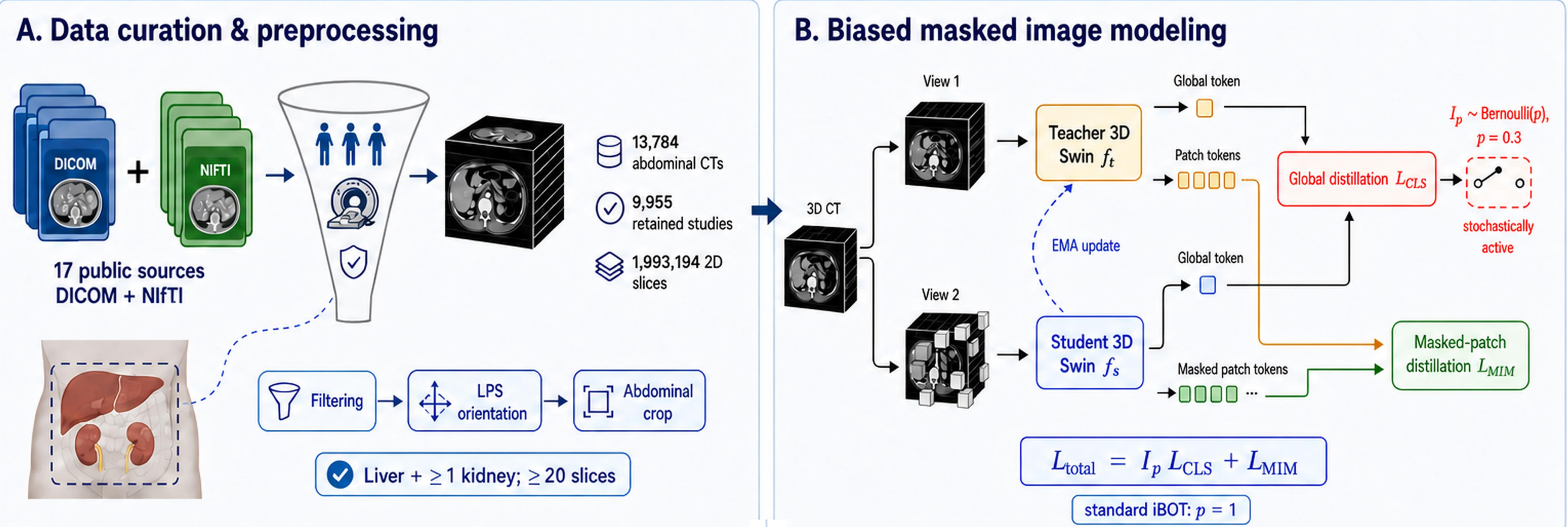
Training
17 public sources
13,784 abdominal CTs
9,955 retained studies
1,993,194 slices

Evaluation

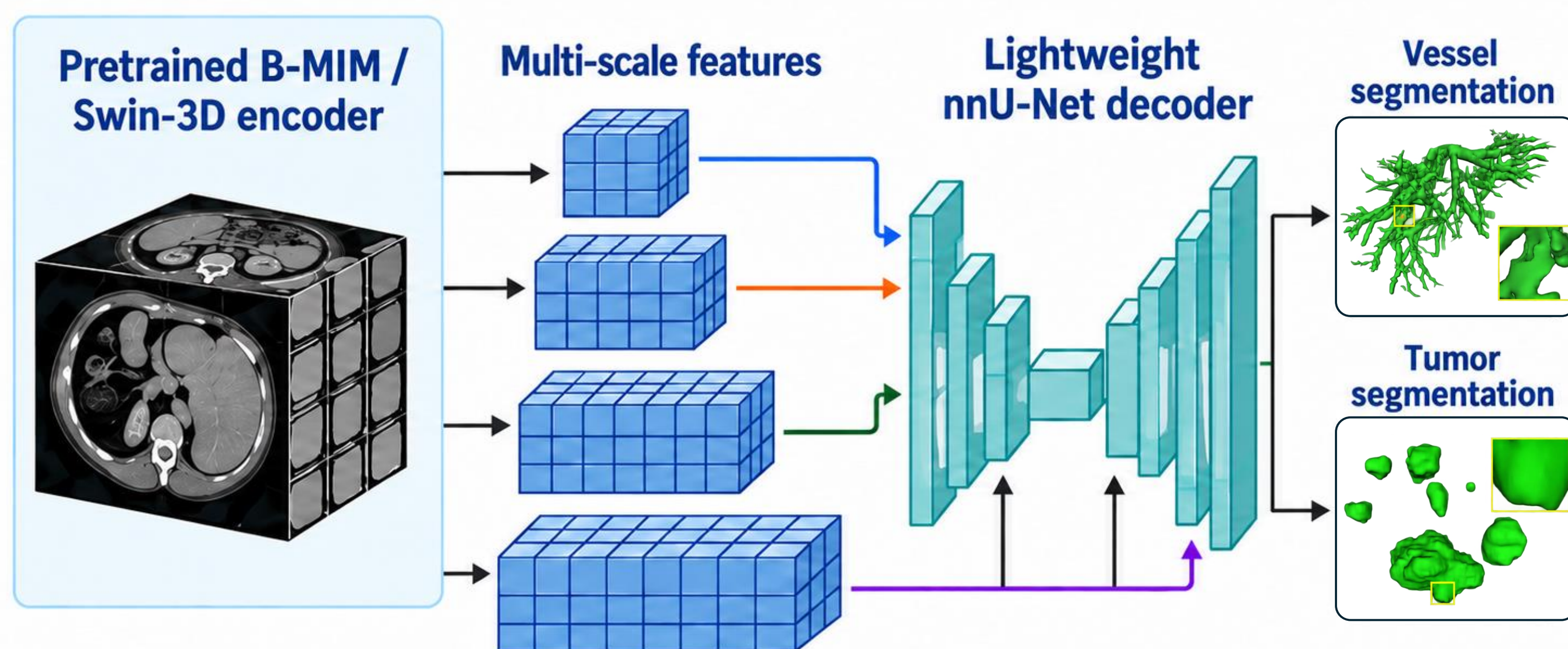
CRLM	IRCAD	MSD
197 studies (liver vessels) 5-fold cross valid	20 studies (liver vessels) Test set	303 studies (small tumors) Test set

2. Method

We reduce global semantic pressure during self-distillation, encouraging the encoder to focus on local high-frequency details.



C. Parameter-efficient downstream adaptation



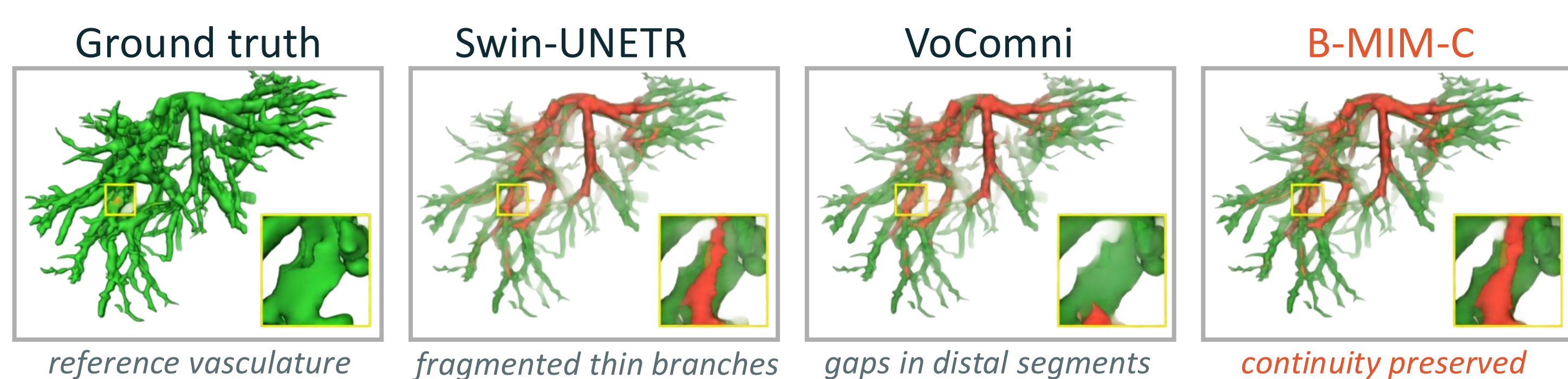
Alternative adaptation strategies

- A Feature concatenation**
 - Largely frozen encoder
 - 16,344 trainable encoder parameters
- B Full fine-tuning**
 - 5,716,410 encoder parameters updated
- C LoRA bottleneck**
 - Frozen backbone
 - Low-rank adapters

4. Results

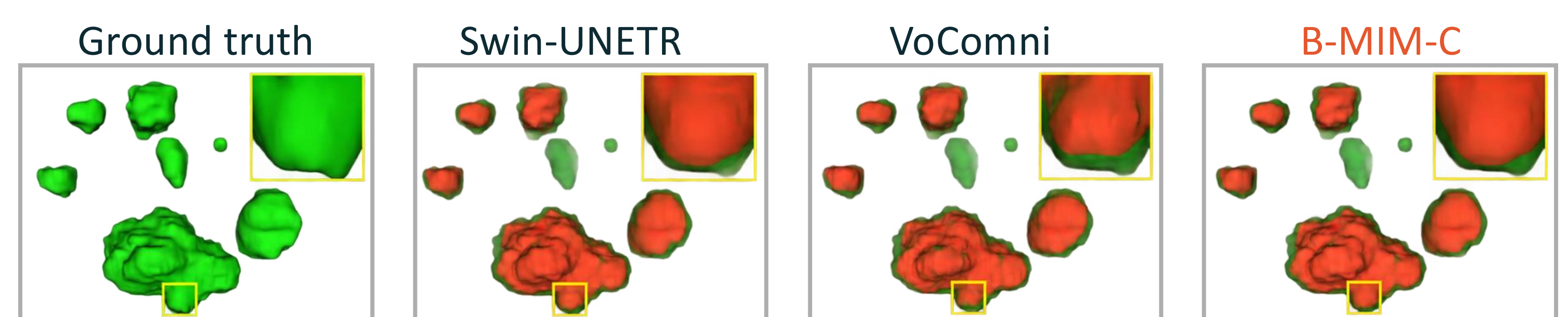
Liver Vessel Segmentation (Train: CRLM, Test: IRCAD and MSD.)

Model	CRLM Dice	CRLM cDice	IRCAD Dice	IRCAD cDice	MSD Dice	MSD cDice
Swin-nnUNet	0.746	0.780	0.400	0.491	0.598	0.624
VoCo_SSL	0.616	0.645	0.382	0.437	0.488	0.480
VoComni	0.632	0.662	0.455	0.494	0.572	0.588
B-MIM-A	0.744	0.777	0.413	0.500	0.611	0.631
B-MIM-B	0.745	0.780	0.401	0.490	0.604	0.630
B-MIM-C	0.746	0.779	0.492	0.582	0.567	0.595

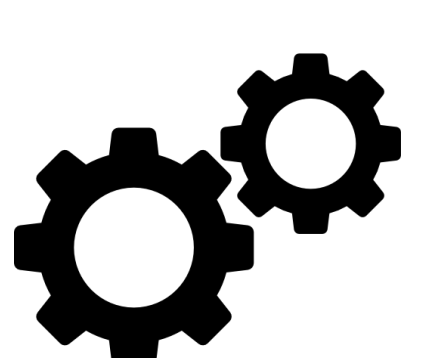


Small Tumor Segmentation (Train: CRLM, Test: IRCAD and MSD.)

Model	CRLM Dice	IRCAD Dice	MSD Dice
Swin-nnUNet	0.646	0.572	0.374
VoCo_SSL	0.650	0.369	0.288
VoComni	0.666	0.500	0.423
B-MIM-A	0.631	0.497	0.357
B-MIM-B	0.651	0.578	0.395
B-MIM-C	0.637	0.506	0.269



5. Key Takeaways



Reducing global semantic pressure improves fine-grained sensitivity.



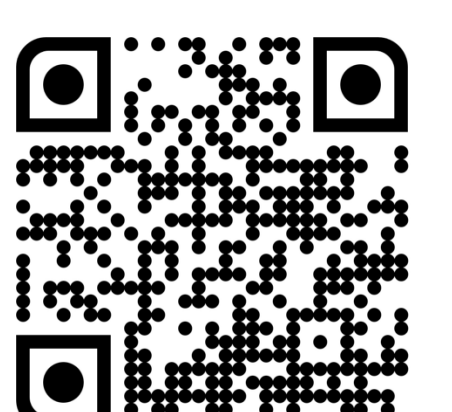
Competitive performance with a fraction of the trainable parameters.



Potential for low-resource settings.



B-MIM is applicable across datasets and tasks



Welcome to connect!

Github repository

